

Business Analysis

BUSINESS IMPACT REPORT

How Data Analysis Transformed APL Logistics' Commercial Decision-Making

APL Logistics (KWE Group) | May 2026

Prepared for: HR, Business Stakeholders & Non-Technical
Leadership

\$8.6M Revenue Analyzed	180K+ Orders Examined	23% Loss-Making Orders Found	6 Business Problems Solved
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Chapter 1: The Problem — Flying Blind on Profit

Imagine running a global logistics company handling over 180,000 deliveries a year, serving customers across five continents, and moving products worth millions of dollars — yet having no clear answer to the most basic business question: "Are we actually making money?"

That was the reality facing APL Logistics. The company had rich data — every order, every customer, and every shipment was recorded in detail. But nobody had connected the dots between all that activity and the profit it was (or wasn't) generating.



The Core Business Gap

APL Logistics could tell you how many orders they shipped. They could tell you total revenue. But they couldn't tell you which customers were profitable, which products were losing money, or whether their discount strategy was helping or hurting the bottom line.

Leadership was making pricing, customer-relationship, and investment decisions, all without knowing which parts of the business were genuinely creating value. This is what data analysts call being "revenue-focused but profit-blind".

What Was at Risk?

- ◆ Sales teams were chasing revenue from customers who were actually costing the company money
- ◆ Discount strategies designed to win orders were quietly destroying profit margins
- ◆ Entire product categories were generating losses that went undetected for months
- ◆ Resources and investment were flowing into markets that delivered the weakest returns



The Mission

This project was commissioned to answer one fundamental question: which customers, products, and markets are truly creating value for APL Logistics, and which ones are not?

Chapter 2: The Investigation — What We Did

Before you can solve a business problem, you need to understand it completely. Our team spent weeks digging through the data — not just looking at surface numbers, but asking the right questions and following the evidence.

We analysed 180,519 order records — every single transaction APL Logistics processed. Think of it like conducting a financial health check on every order the company ever made, one by one.

Our Four Investigation Areas



Area 1: Customer Profitability

We looked at every customer individually and asked, after all costs and discounts, is this customer actually making us money? The answer was surprising; many high-volume customers were generating more costs than profit.



Area 2: Product & Category Performance

We examined every product category to find which ones had healthy profit margins and which ones were secretly losing money despite generating strong sales numbers.



Area 3: Discount Strategy

We studied the relationship between discounts given to customers and the profit generated from those orders. We were looking for the "tipping point", the discount level beyond which the business stops making money.



Area 4: Market Performance

We compared all five global markets — North America, Europe, Latin America, Asia Pacific, and Africa — to understand which ones were genuinely profitable and which were at risk.

How Long Did It Take?

Phase	What We Did	Time
Data Preparation	Cleaned and validated all 180,519 records	Week 1
Customer Analysis	Examined every customer's true profit contribution	Week 2
Product Analysis	Calculated real margins for every category	Week 2
Discount Analysis	Found the profit tipping point	Week 3
Market Analysis	Compared all 5 global markets	Week 3
Dashboard Build	Built live interactive reporting tool	Week 4

Chapter 3: The Discoveries — What We Found

Some findings confirmed what leadership suspected. Others were genuinely shocking. Here is what the data revealed, told as plainly as possible.

Discovery 1: Nearly 1 in 4 Orders Lost Money

This was the most alarming headline finding. Of the 180,519 orders we analysed, approximately 23% — nearly one in every four — generated a negative profit. That means the company was spending more to fulfil those orders than it was earning from them.



What This Means in Plain English

Imagine a shop where 23% of customers walk out having paid less than the cost of their items. The shop looks busy, and the till is ringing – but it's quietly losing money on a quarter of all transactions. That was APL Logistics.

Discovery 2: Some Products Were Destroying Value

When we broke down profitability by product category, we found a dramatic split between winners and losers:

Category	What Was Happening	Business Impact
Golf Bags & Carts	✓ For every \$1 earned, 24 cents was profit	High value – grow this
Basketball	✓ 18 cents profit per dollar	Strong performer – invest more
Lacrosse	✓ Nearly 20 cents profit per dollar	Underserved opportunity
Electronics	⚠ Only 1 cent profit per dollar on \$1M+ revenue	Huge revenue, almost no return
As Seen on TV	✗ Losing money on every sale	Small but loss-making
Strength Training	✗ Losing 24 cents on every \$1 of sales	Critical – stop or reprice immediately



The Insight Nobody Expected

Electronics was the highest-revenue category – over \$1 million in sales. But it was generating almost zero profit (just 1.1%). The company was working incredibly hard for almost no financial return in its biggest category.

Discovery 3: The Discount Danger Zone

This discovery directly translates into a simple, actionable rule that any manager can understand and apply:

Discount Given	Profit Result	What It Means
0% – 5%	Healthy profit	✅ Safe zone — encourage this
5% – 10%	Healthy profit	✅ Still good — acceptable
10% – 20%	Stable profit	✅ Monitor closely
20% – 30%	Shrinking profit	⚠️ Risky — limit usage
Above 30%	ZERO profit	❌ Hard stop — money is gone



The Single Most Valuable Finding

When discounts exceed 30%, the average profit on those orders drops to zero. The business is essentially giving products away. This finding alone, if acted upon, could recover millions in lost profit.

Discovery 4: Market Performance Varied Dramatically

Not all markets were created equal. North America (called USCA — US and Canada) was the strongest market. Africa and Latin America showed concerning signs of margin pressure that, if left unaddressed, could tip those markets into loss-making territory.

Chapter 4: The Solution — What We Built

Findings alone don't change businesses. We needed to make this information permanently accessible to the people who make decisions at APL Logistics — without requiring them to be data experts.

So we built a live, interactive dashboard that any manager can open in their browser and use immediately. No technical training required. No data science degree needed. Just click, filter, and see the answers.



The APL Logistics Profitability Dashboard

A live web application that shows revenue, profit, customer value, product margins, discount impact, and market performance – all in one place, all updating in real time as you apply filters.

What Can a Manager Do With It?

- ◆ Ask, "Which customers should we prioritise this quarter?" and get an instant ranked list by profitability
- ◆ Ask, "What happens to our profit if we increase discounts by 5%?" and see a live simulation
- ◆ Ask "which product categories should we grow vs. stop selling?" and see margin rankings instantly
- ◆ Ask "is our Latin America market profitable?" and see a real-time margin comparison



Accessible to Everyone

The dashboard was specifically designed so that non-technical users — including HR, finance, and commercial leadership — can navigate it without any data training. If you can use a website, you can use this dashboard.

Chapter 5: The Impact — Problems We Solved

This project didn't just produce a report. It changed how APL Logistics can think about its business. Here is a plain-English summary of each problem we solved.

Business Problem	Before This Project	After This Project
Which customers are profitable?	No visibility — managed by revenue only	Clear profit ranking of every customer
Which products make money?	Assumed all categories were positive	Identified 2 loss-making categories immediately
Is our discount strategy working?	No data — discounts given freely	Hard 30% cap identified — policy can change tomorrow
Which markets to invest in?	Based on revenue size only	Profit margin ranking of all 5 markets
Are we growing profitably?	Revenue growth only tracked	Profit margin tracked alongside revenue
Who are our best customers?	Whoever buys the most	Whoever generates the most profit after costs

The Financial Opportunity



Potential Margin Improvement

If APL Logistics implements the six recommended actions from this analysis, the business could realistically improve its overall profit margin from 11.3% to 15–18% within 12 months. On \$8.6M in revenue, that represents hundreds of thousands of dollars in additional profit—without winning a single new customer.

Chapter 6: What We Recommend — The Six Actions

Data analysis is only valuable if it leads to action. Here are the six recommendations from this project, written so any business leader can understand and act on them immediately.

Sr. no.	The Action	Why It Matters	When
1	Stop approving discounts above 30%	Discounts above 30% generate zero profit. Every such order costs us money.	This week
2	Stop selling Strength Training at current prices	We lose 24 cents on every dollar. It is not commercially viable.	This month
3	Review Electronics pricing urgently	We earn only 1 cent profit per dollar on our biggest revenue category.	60 days
4	Invest more in Golf, Basketball & lacrosse.	These categories earn 18–24 cents profit per dollar — 10x better.	90 days
5	Segment customers into value tiers	Focus service and relationship investment on customers who generate profit.	90 days
6	Apply tighter discount rules in Africa & LATAM	These markets have the thinnest margins and highest loss risk.	60 days

In Closing: What This Project Proves

This project began with a simple but powerful question: Does APL Logistics know which parts of its business are truly profitable?

The answer, before this analysis, was **no**. The answer now is **yes**.

Through careful examination of 180,519 orders, we found that the company's commercial reality was significantly different from what revenue figures suggested. Some of its most active product categories were losing money. A significant share of its orders was unprofitable. And a simple change to its discount policy could recover substantial profit without any operational change.



The Core Lesson

This project demonstrates that data analysis is not about technology or complexity. It is about asking the right business questions and having the discipline to follow the evidence – wherever it leads. APL Logistics now has both the answers and the tools to act on them.

The live dashboard built as part of this project ensures these insights are not a one-time report; they are permanently embedded into how APL Logistics monitors and manages its commercial performance going forward.

— End of Business Impact Report —

APL Logistics (KWE Group) | Unified Mentor Program | May 2026 | Confidential

Data Analysis

TECHNICAL ANALYSIS REPORT

Customer, Product & Profitability Performance Analysis in Supply Chain Operations

APL Logistics (KWE Group) | Unified Mentor Program | May 2026

Tools: Python · Pandas · Plotly · Streamlit · Google Colab · GitHub

1. Project Overview

This technical report documents the end-to-end data analysis pipeline developed for APL Logistics, a global supply chain provider under the KWE Group. The objective was to build a profitability intelligence system that transforms raw transactional data into actionable commercial insights across customers, products, categories, and markets.

TECH STACK

Layer	Tool	Purpose
Data Analysis	Python + Pandas	EDA, cleaning, aggregation
Visualization	Plotly Express / Graph Objects	Interactive charts & maps
Dashboard	Streamlit	Live web application
Environment	Google Colab	Cloud-based notebook EDA
Version Control	GitHub	Code repository & deployment
Deployment	Streamlit Cloud	Public dashboard hosting

2. Dataset Overview

2.1 Dataset Specifications

Property	Value
Total Records	180,519 rows
Total Features	40 columns
Null Values	0 (fully clean)
Duplicate Rows	0
File Size	~60MB CSV
Time Period	Multi-year transactional data
Source	APL Logistics Order Management System

2.2 Key Variable Categories

- ▶ **Financial:** Sales, Order Profit Per Order, Benefit Per Order, Order Item Discount Rate, Order Item Profit Ratio
- ▶ **Customer:** Customer ID, Customer Segment (Consumer/Corporate/Home Office), Geography fields
- ▶ **Product:** Product Name, Product Price, Category Name, Department Name
- ▶ **Operational:** Shipping Mode, Delivery Status, Late Delivery Risk, Days for Shipping
- ▶ **Geographic:** Market, Order Region, Order Country, Latitude, Longitude

3. Exploratory Data Analysis (EDA)

3.1 Environment Setup — Google Colab

All EDA was conducted in Google Colab using Python 3. The following libraries were imported:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import warnings
warnings.filterwarnings('ignore')
```

SCREENSHOT: COLAB ENVIRONMENT & IMPORTS

	Type	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	Delivery Status	Late_delivery_risk	Category Id	Category Name	Customer City	...	Order Item Quantity	Sales	Order Item Total	Order Profit Per Order	Order Region	Order State	Order Status	Product Name	Product Price	Shipping Mode
0	DEBIT	6	4	159.69	472.45	Late delivery	1	9	Cardio Equipment	Brownsville	...	5.0	499.95	472.45	159.69	South Asia	Maharashtra	COMPLETE	Nike Men's Free 5.0+ Running Shoe	99.99	Standard Class
1	DEBIT	4	4	48.71	167.96	Shipping on time	0	29	Shop By Sport	Littleton	...	5.0	199.95	167.96	48.71	Central America	Cortés	ON_HOLD	Under Armour Girls' Toddler Spine Surge Runner	39.99	Standard Class
2	DEBIT	4	4	87.36	181.99	Shipping on time	0	48	Water Sports	Littleton	...	1.0	199.99	181.99	87.36	Central America	Cortés	ON_HOLD	Pelican Sunstream 100 Kayak	199.99	Standard Class
3	DEBIT	6	4	-41.89	175.99	Late delivery	1	48	Water Sports	Littleton	...	1.0	199.99	175.99	-41.89	East of USA	Nueva York	COMPLETE	Pelican Sunstream 100 Kayak	199.99	Standard Class
4	DEBIT	6	4	10.00	40.00	Late delivery	1	24	Women's Apparel	Littleton	...	1.0	50.00	40.00	10.00	East of USA	Nueva York	COMPLETE	Nike Men's Dri-FIT Victory Golf Polo	50.00	Standard Class

5 rows x 40 columns

3.2 Data Loading & Initial Inspection

The dataset was loaded and inspected for shape, data types, null values, and duplicate records:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 43984 entries, 0 to 43983
Data columns (total 40 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Type                                  43984 non-null  object
1   Days for shipping (real)              43984 non-null  int64
2   Days for shipment (scheduled)        43984 non-null  int64
3   Benefit per order                    43984 non-null  float64
4   Sales per customer                   43984 non-null  float64
5   Delivery Status                      43984 non-null  object
6   Late_delivery_risk                  43984 non-null  int64
7   Category Id                         43984 non-null  int64
8   Category Name                       43984 non-null  object
9   Customer City                       43984 non-null  object
10  Customer Country                    43984 non-null  object
11  Customer Fname                      43984 non-null  object
12  Customer Id                         43983 non-null  float64
13  Customer Lname                      43983 non-null  object
14  Customer Segment                    43983 non-null  object
15  Customer State                      43983 non-null  object
16  Customer Street                     43983 non-null  object
17  Customer Zipcode                    43983 non-null  float64
18  Department Id                       43983 non-null  float64
19  Department Name                     43983 non-null  object
20  Latitude                            43983 non-null  float64
21  Longitude                           43983 non-null  float64
22  Market                              43983 non-null  object
23  Order City                          43983 non-null  object
24  Order Country                       43983 non-null  object
25  Order Customer Id                   43983 non-null  float64
26  Order Item Discount                 43983 non-null  float64
27  Order Item Discount Rate            43983 non-null  float64
28  Order Item Product Price            43983 non-null  float64
29  Order Item Profit Ratio             43983 non-null  float64
30  Order Item Quantity                 43983 non-null  float64
31  Sales                              43983 non-null  float64
32  Order Item Total                    43983 non-null  float64
33  Order Profit Per Order              43983 non-null  float64
34  Order Region                        43983 non-null  object
35  Order State                         43983 non-null  object
36  Order Status                        43983 non-null  object
37  Product Name                        43983 non-null  object
38  Product Price                       43983 non-null  float64
39  Shipping Mode                       43983 non-null  object
dtypes: float64(17), int64(4), object(19)
memory usage: 13.4+ MB
```

3.3 Step-by-Step EDA Methodology

Step 1 — Statistical Summary

```
df.describe()
```

Generated descriptive statistics for all 40 columns. Key observations: Sales ranged from near-zero to high values; profit ratios showed significant variance, indicating inconsistent margin performance.

	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	late_delivery_risk	Category Id	Customer Id	Customer Zipcode	Department Id	Latitude	...	Order Customer Id	Order Item Discount	Order Item Discount Rate	Order Item Product Price	Order Item Profit Ratio	Order Item Quantity	Sales
count	43984.000000	43984.000000	43984.000000	43984.000000	43984.000000	43984.000000	43983.000000	43983.000000	43983.000000	43983.000000	—	43983.000000	43983.000000	43983.000000	43983.000000	43983.000000	43983.000000	43983.000000
mean	3.483881	2.931725	21.404170	179.499008	0.543470	30.016870	1586.571675	36479.505036	5.338085	29.508490	—	1586.571675	20.294303	0.101744	133.821258	0.121005	2.186004	199.794348
std	1.616921	1.371843	100.195976	103.126206	0.496112	13.724129	910.752295	37739.357686	1.482613	9.653778	—	910.752295	19.888379	0.070403	119.075124	0.466378	1.469107	113.464894
min	0.000000	0.000000	-4274.980000	7.450000	0.000000	2.000000	1.000000	603.000000	2.000000	17.982491	—	1.000000	0.000000	0.000000	9.990000	-2.750000	1.000000	9.990000
25%	2.000000	2.000000	7.430000	106.590000	0.000000	18.000000	791.000000	725.000000	4.000000	18.261364	—	791.000000	5.500000	0.040000	50.000000	0.080000	1.000000	119.980000
50%	3.000000	4.000000	31.870000	163.990000	1.000000	26.000000	1597.000000	19720.000000	5.000000	32.965931	—	1597.000000	14.400000	0.100000	59.990000	0.270000	1.000000	199.920000
75%	5.000000	4.000000	63.900000	245.960000	1.000000	45.000000	2366.000000	76520.000000	7.000000	38.604679	—	2366.000000	29.990000	0.160000	199.990000	0.360000	3.000000	299.950000
max	6.000000	4.000000	864.000000	1899.990000	1.000000	48.000000	3180.000000	98632.000000	7.000000	47.912502	—	3180.000000	360.000000	0.250000	1999.990000	0.500000	5.000000	1999.990000

8 rows x 21 columns

Step 2 — Revenue & Profit KPIs

Revenue & Profit Overview (KPIs)

```
total_revenue = df['Sales'].sum()
total_profit = df['Order Profit Per Order'].sum()
profit_margin = (total_profit / total_revenue) * 100
avg_discount = df['Order Item Discount Rate'].mean() * 100

print(f"Total Revenue:    ${total_revenue:,.2f}")
print(f"Total Profit:       ${total_profit:,.2f}")
print(f"Profit Margin:        {profit_margin:,.2f}%")
print(f"Avg Discount:        {avg_discount:,.2f}%")
```

Total Revenue:

\$8,787,554.82

Total Profit:

\$941,428.68

Profit Margin:

10.71%

Avg Discount:

10.17%

Step 3 — Customer Analysis

```
▶ customer_df = df.groupby('Customer Id').agg(  
    Total_Sales = ('Sales', 'sum'),  
    Total_Profit = ('Order Profit Per Order', 'sum'),  
    Order_Count = ('Sales', 'count')  
) .reset_index()  
  
customer_df['Profit_Margin'] = (customer_df['Total_Profit'] / customer_df['Total_Sales']) * 100  
  
# Top 10 most profitable customers  
print(customer_df.sort_values('Total_Profit', ascending=False).head(10))  
  
# Loss-making customers  
print("Loss-making customers:", (customer_df['Total_Profit'] < 0).sum())
```

***	Customer Id	Total_Sales	Total_Profit	Order_Count	Profit_Margin	
	2612	2641.0	9130.92	2441.97	43	26.743964
	1636	1657.0	9223.71	2196.92	42	23.818182
	2597	2626.0	6274.36	1928.57	27	30.737318
	741	749.0	7649.38	1855.15	38	24.252292
	2198	2224.0	6774.46	1799.87	31	26.568464
	1133	1148.0	6524.42	1716.96	22	26.315902
	2768	2800.0	5443.48	1680.45	27	30.870877
	2691	2723.0	7719.08	1674.83	30	21.697275
	475	482.0	6689.26	1668.63	34	24.944912
	2154	2180.0	5969.44	1623.26	27	27.192836
	Loss-making customers: 660					

Step 4 — Product & Category Profitability

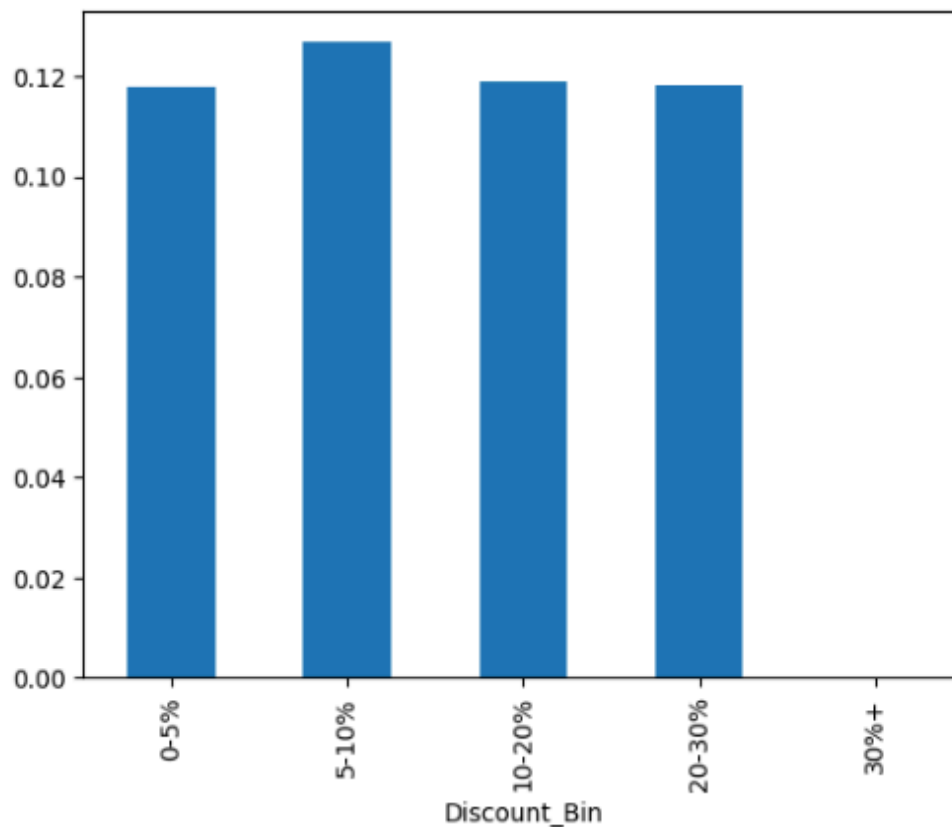
```
category_df = df.groupby('Category Name').agg(  
    Revenue = ('Sales', 'sum'),  
    Profit = ('Order Profit Per Order', 'sum')  
).reset_index()  
category_df['Margin_%'] = (category_df['Profit'] / category_df['Revenue']) * 100  
print(category_df.sort_values('Margin_%', ascending=False))
```

```
...      Category Name  Revenue  Profit  Margin_%  
13      Golf Bags & Carts    1869.89    461.51  24.681131  
21      Lacrosse    10172.34    2004.49  19.705299  
3      Basketball    8199.82    1468.34  17.906978  
2      Baseball & Softball  25995.15    4038.80  15.536744  
25      Soccer    8749.03    1292.96  14.778324  
31      Women's Golf Clubs  12898.79    1884.17  14.607339  
20      Kids' Golf Clubs  26669.20    3828.77  14.356524  
28      Trade-In    17230.98    2247.88  13.045573  
0      Accessories    32187.12    4165.36  12.941077  
23      Men's Golf Clubs  12553.85    1559.63  12.423519  
12      Golf Apparel    9337.27    1156.62  12.387132  
27      Tennis & Racquet    9807.82    1168.25  11.911414  
30      Women's Apparel  823750.00    93565.31  11.358459  
22      Men's Footwear  733533.57    82290.43  11.218359  
8      Electronics    102268.97    11384.17  11.131597  
24      Shop By Sport  335558.74    36980.16  11.020473  
9      Fishing    1750712.46    188301.86  10.755727  
6      Cardio Equipment  958295.01    102683.88  10.715268  
7      Cleats    1134771.24    120515.27  10.620226  
19      Indoor/Outdoor Games  708666.42    75105.69  10.598173  
10      Fitness Accessories    10127.59    1073.26  10.597388  
5      Camping & Hiking  1058629.42    111284.56  10.512136  
29      Water Sports    808859.63    84433.39  10.438571  
11      Girls' Apparel    38617.86    3986.23  10.322245  
14      Golf Balls    19271.00    1873.91  9.723989  
16      Golf Shoes    27196.00    2240.01  8.236542  
15      Golf Gloves    30767.60    2467.20  8.018825  
18      Hunting & Shooting  14175.43    1117.83  7.885687  
17      Hockey    12320.94    789.15  6.404950  
4      Boxing & MMA    21182.39    463.94  2.190216  
1      As Seen on TV!    6199.38    -309.82  -4.997597  
26      Strength Training  16979.91    -4094.53  -24.113968
```

Step 5 — Discount Impact Analysis

```
# Bin discount rates and see effect on profit ratio  
df['Discount_Bin'] = pd.cut(df['Order Item Discount Rate'],  
    bins=[0, 0.05, 0.1, 0.2, 0.3, 1.0],  
    labels=['0-5%', '5-10%', '10-20%', '20-30%', '30%+'])  
  
df.groupby('Discount_Bin')['Order Item Profit Ratio'].mean().plot(kind='bar')
```

SCREENSHOT: DISCOUNT IMPACT BAR CHART



Step 6 — Regional & Market Analysis

```
market_df = df.groupby('Market').agg(  
    Revenue = ('Sales', 'sum'),  
    Profit = ('Order Profit Per Order', 'sum')  
) .reset_index()  
market_df['Margin_%'] = (market_df['Profit'] / market_df['Revenue']) * 100  
print(market_df.sort_values('Margin_%', ascending=False))
```

	Market	Revenue	Profit	Margin_%
2	LATAM	2614897.23	307028.30	11.741505
4	USCA	1284600.52	139805.79	10.883211
0	Africa	609087.68	63236.94	10.382239
1	Europe	2529071.75	256506.09	10.142302
3	Pacific Asia	1749897.64	174851.56	9.992102

4. Streamlit Dashboard Architecture

4.1 Application Structure

Module	Charts Included	Key Insight Delivered
Revenue & Profit Overview	Grouped bar, pie, horizontal bar	Market-level profit vs revenue gap
Customer Value Dashboard	Bar charts, pie	Top/bottom 15 customers, tier distribution
Product & Category	Margin bar, bubble chart	PUSH vs DROP categories
Discount Analyzer	Scatter, bar, waterfall	30% threshold finding
Regional Map	Treemap, geo scatter	Global margin heat map

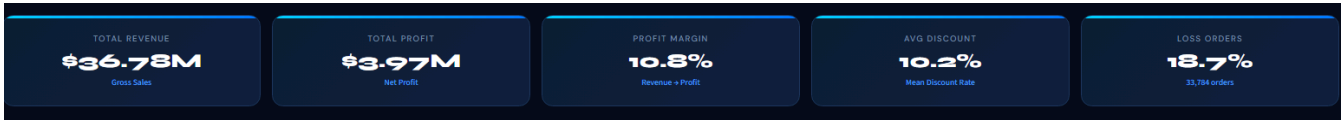
4.2 Sidebar Filter System

- ▶ Customer Segment dropdown (Consumer / Corporate / Home Office)
- ▶ Market dropdown (USCA / Europe / LATAM / APAC / Africa)
- ▶ Product Category selector
- ▶ Shipping Mode filter
- ▶ Max Discount Rate slider (0–100%)

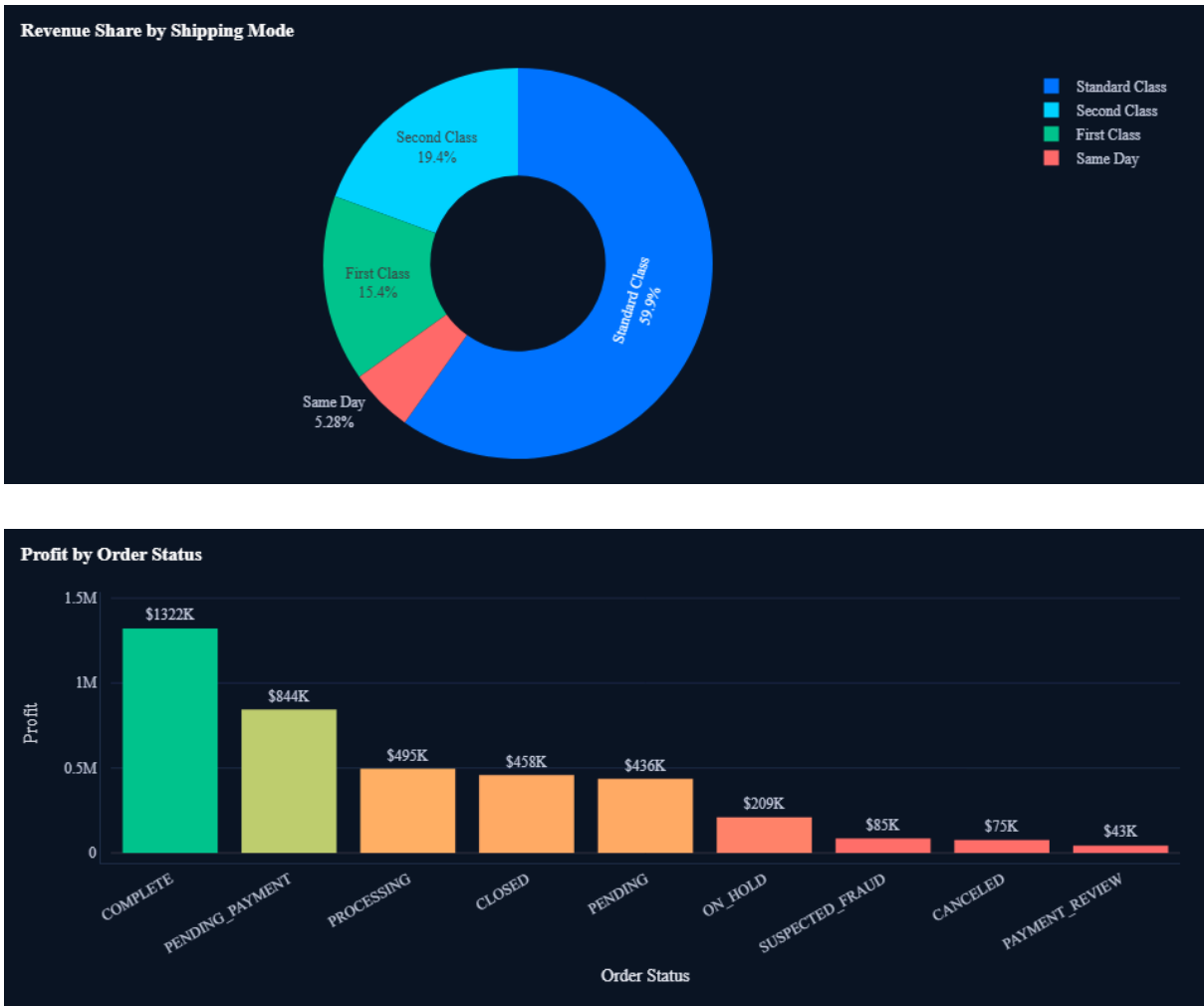
All 5 dashboard modules react dynamically to sidebar filter changes using Streamlit's reactive data flow.

4.3 Dashboard Screenshots

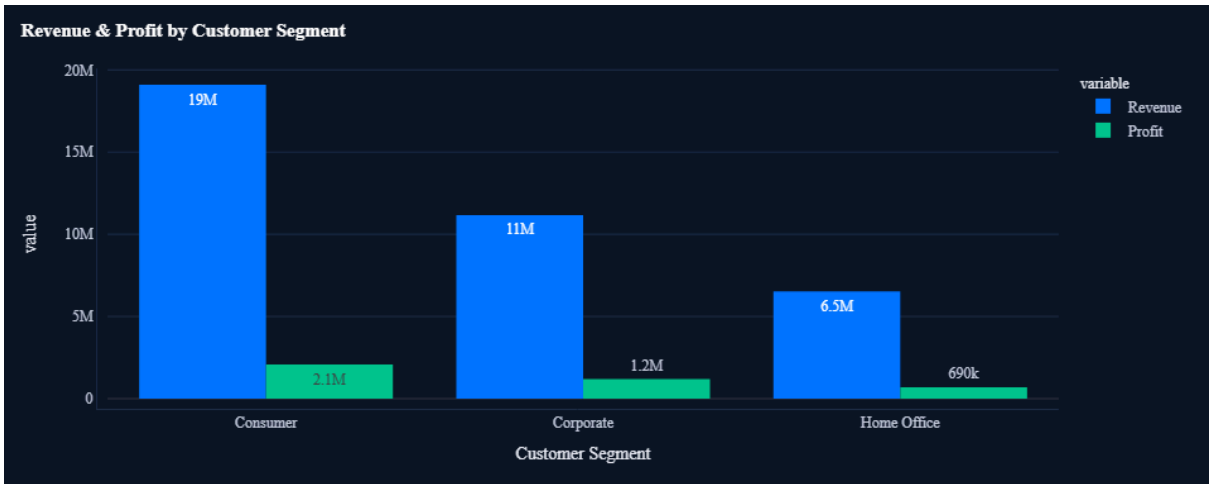
SCREENSHOT 1: DASHBOARD HEADER & KPI CARDS

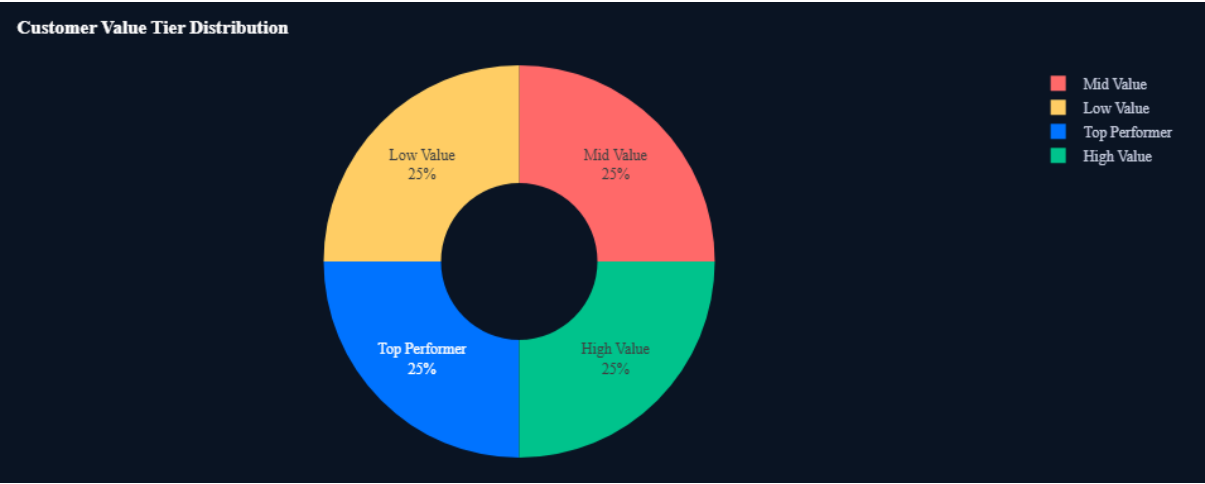


SCREENSHOT 2: REVENUE & PROFIT OVERVIEW MODULE

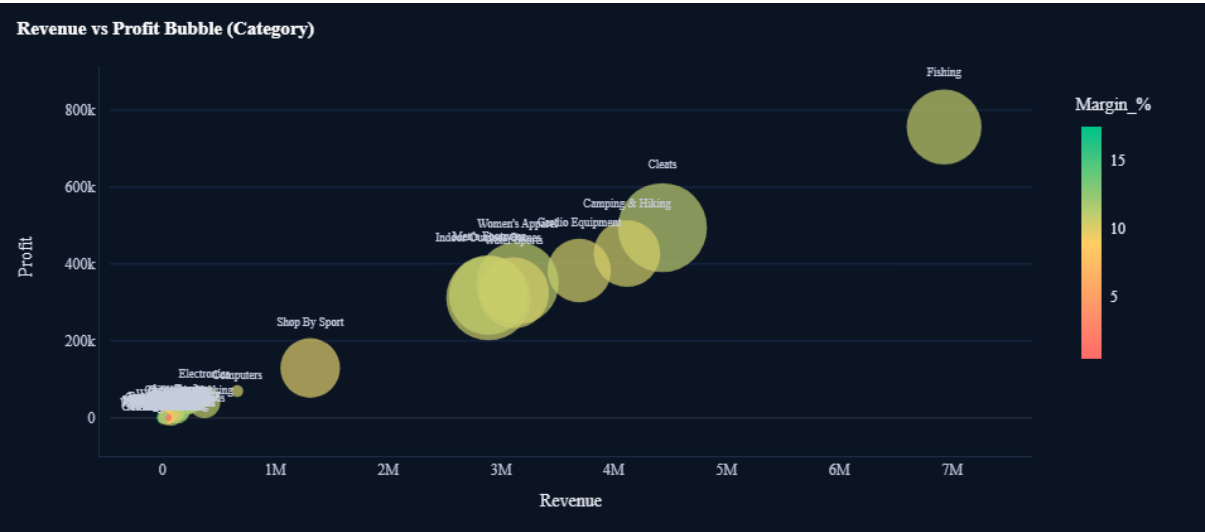
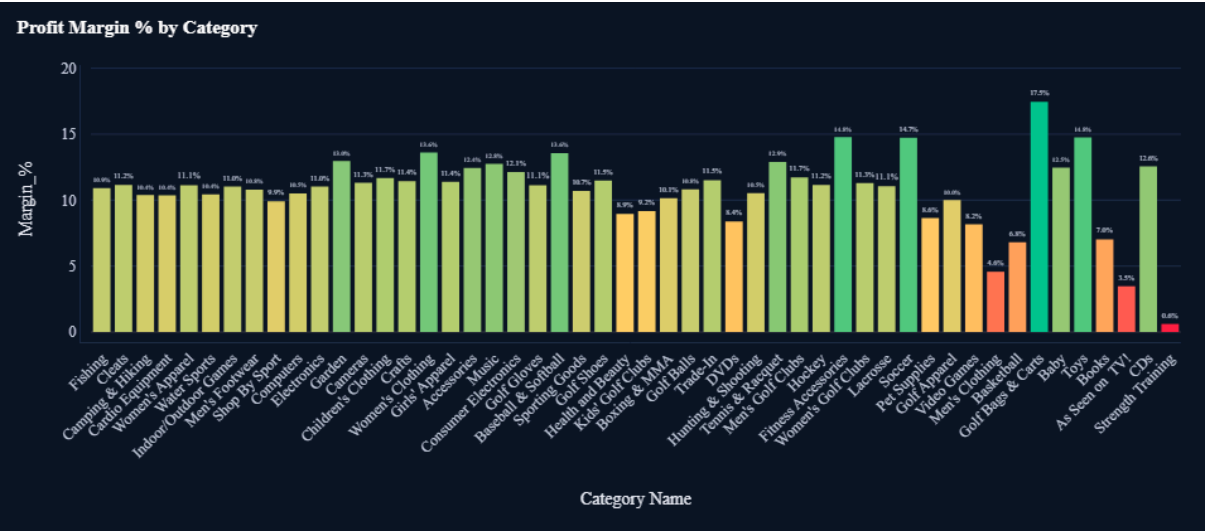


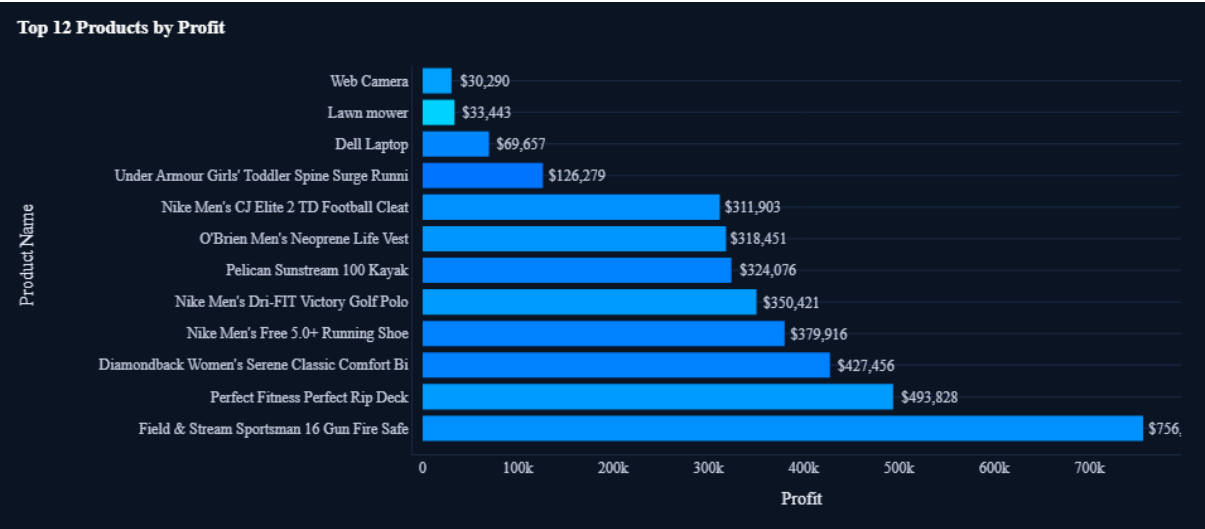
SCREENSHOT 3: CUSTOMER VALUE DASHBOARD



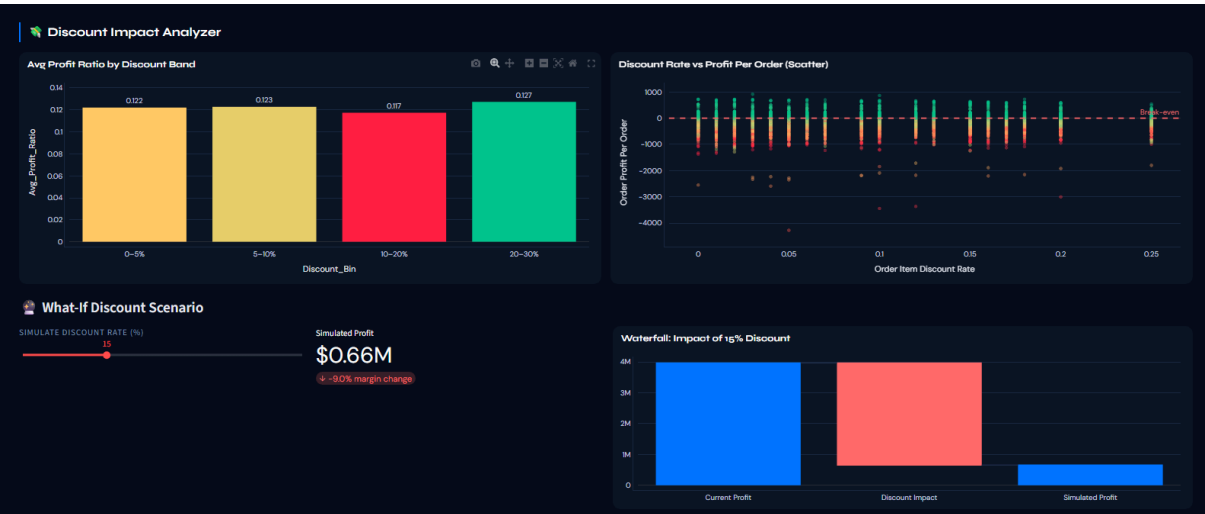


SCREENSHOT 4: PRODUCT & CATEGORY PERFORMANCE

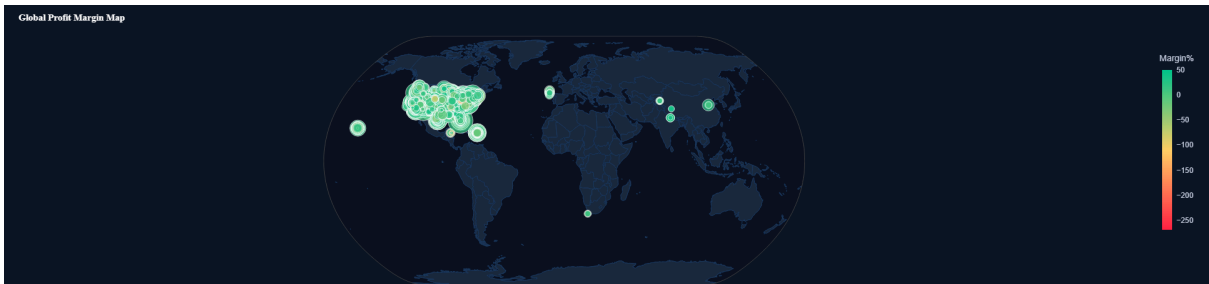
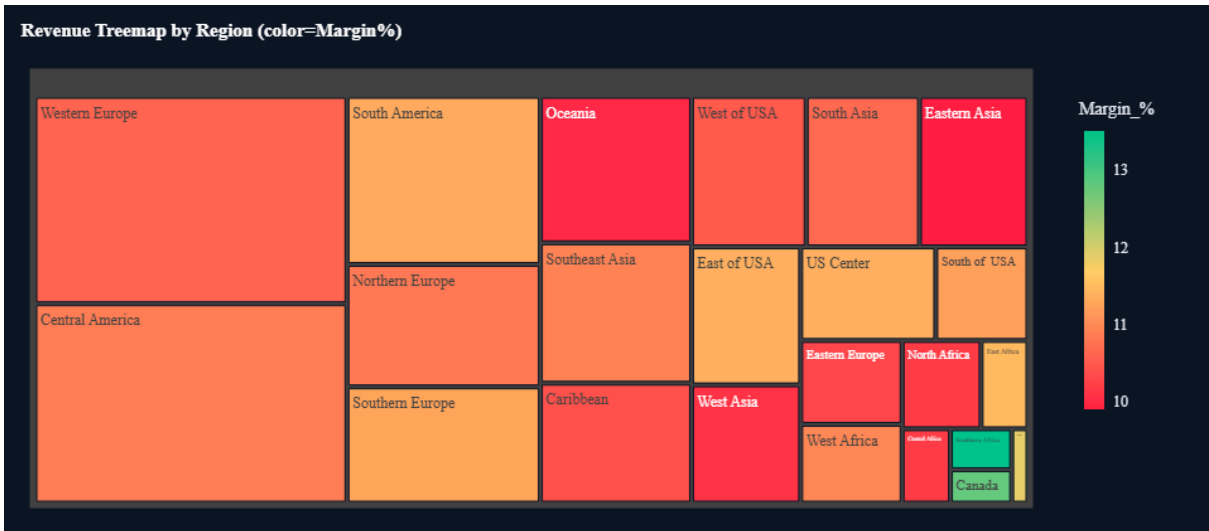
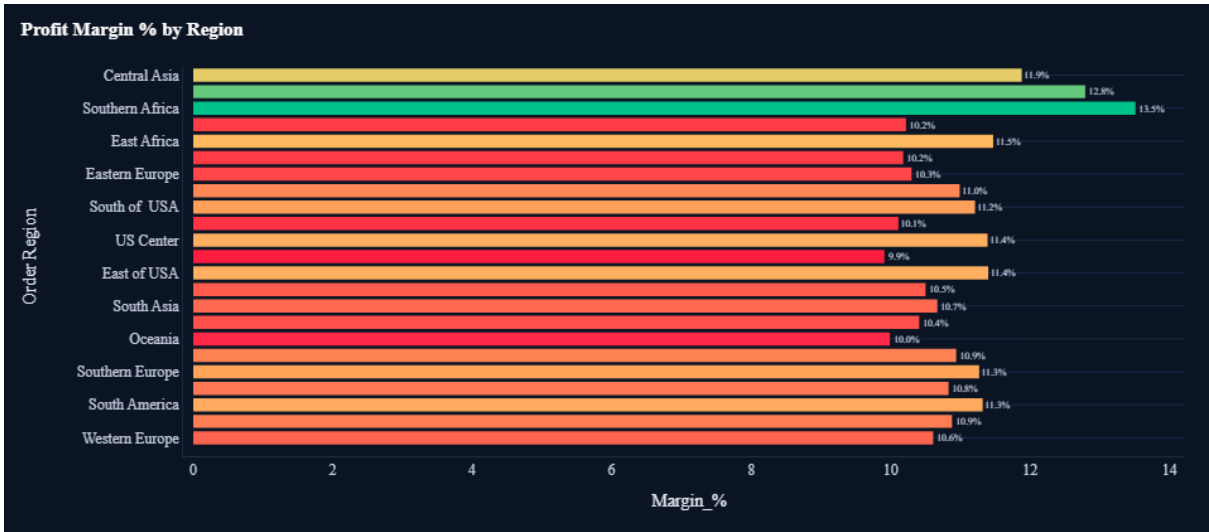




SCREENSHOT 5: DISCOUNT IMPACT ANALYZER



SCREENSHOT 6: GLOBAL MARKET MAP



5. Key Technical Findings

Finding	Method Used	Result
Category Margin Analysis	groupby + margin calc	Strength Training: -24.1%, Golf: +24.0%
Discount Threshold	pd.cut() binning	30%+ discount → 0.00 profit ratio
Loss-Making Orders	Boolean filter	23% of 180,519 orders = negative profit
Top Customer Value	Customer groupby	High profit concentration in top 25%
Market Weakness	Market groupby	Africa & LATAM below 8% margin
Electronics Risk	\$1M+ revenue at 1.1% margin	Highest revenue, near-zero return

6. Deployment Pipeline

Stage	Tool	Detail
Development	VS Code + Python 3.12	Local Streamlit development & testing
Version Control	GitHub	Repository: bigbunny11/Customer-Product-...
Data Hosting	Google Drive	60MB CSV hosted with public access URL
Deployment	Streamlit Cloud	Auto-deploy on GitHub commit
Live URL	pca5ee5gnxuthwebztynyt.streamlit.app	Public dashboard link

— End of Technical Report —